Scientific Reviewer

8+3 · seat empty

Dual-system thesis · Table 2/4 mismatch

Architectural foil to π MoE

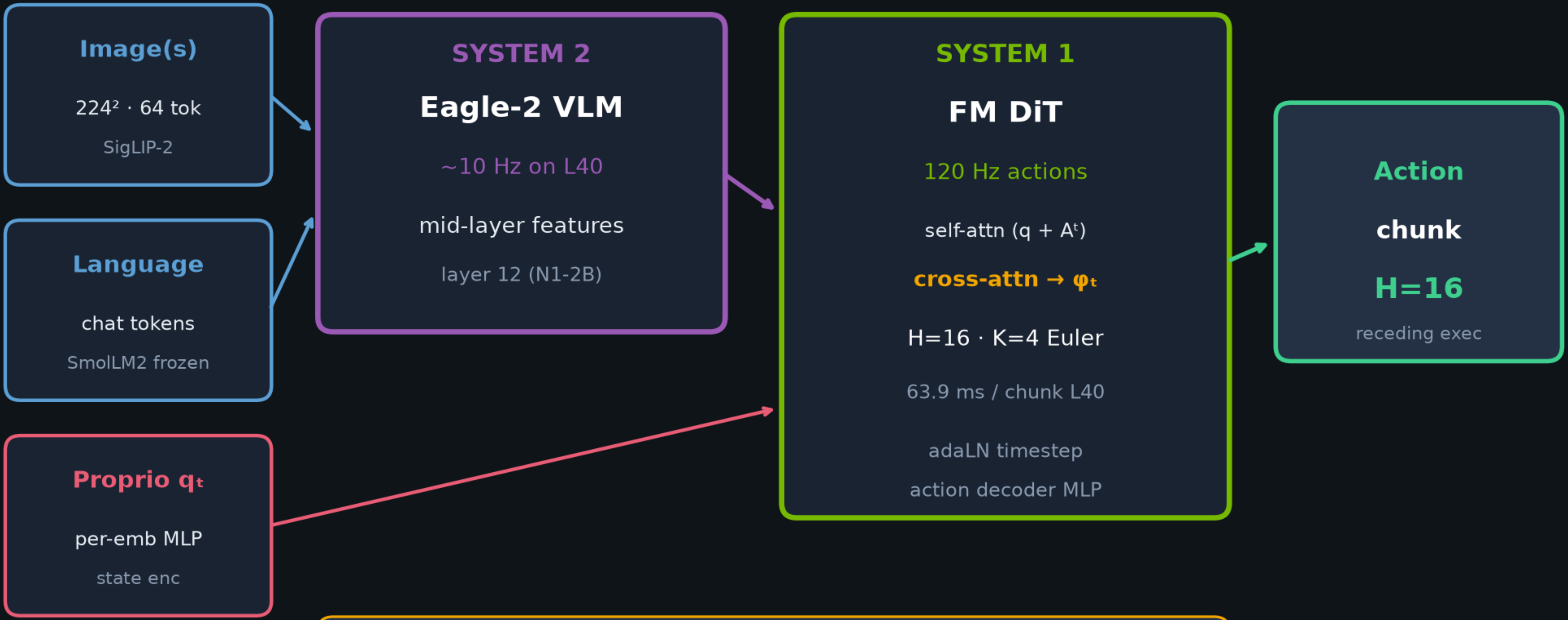
Spine: Eagle-2 System 2 (~10 Hz) + FM DiT System 1 (120 Hz) via cross-attn

H=16 · K=4 · N1 ≠ N1.7 · NOT a classical planner

NVIDIA · Fourier GR-1 · open humanoid VLA

Dual-system VLA · Eagle-2 System 2 × FM DiT System 1

Jointly trained end-to-end · cross-attn bridge · NOT a classical planner



Cross-attn handoff = inspectable System-2 interface · foil to π fused MoE expert

Strengths · falsifiable dual-system thesis

Clear systems thesis	Kahneman dual-system + Hz split + cross-attn vs π MoE
Data methodology 1st-class	LAPA · IDM · DexMimicGen · neural I2V — not just more OXE
Broad eval	3 sim suites × scales + real GR-1 + pretrain probes
Honest forgetting fail	Right-hand-only FT erases handover — useful negative
Open release intent	LeRobot-compatible formats · reproduction path

Weaknesses / threats to validity

Table 2/4 DexMG mismatch

66.5% vs 58.5% @100 — preserve BOTH; needs erratum

IL not VLA baselines

DP / BC-T main tables — not π_0 / OpenVLA

Small-n partial credit

n=10 (n=5 packing); variance underspecified

Confounded ablations

Arch + data + capacity differ from DP U-Net

Short-horizon rhetoric

Tabletop vs “generalist humanoid” claim

Neural gains preliminary

Physics fidelity / MLLM-judge selection bias

Tattoo numbers · remember these

Table 2 sim @100 demos · Table 3 real GR-1 · preserve DexMG mismatch

SIM avg @100 (Table 2)

N1-2B ~45%

vs DP 33.4% · BC-T 26.4%

RoboCasa 32.1 · DexMG 66.5 · GR-1 50.0

App Table 4 DexMG@100 = 58.5 ≠ 66.5

REAL GR-1 avg (Table 3)

N1 full 76.8%

vs DP full 46.4%

N1 @10% → 42.6% (≈ DP full −3.8 pp)

n≈10 trials · Pack Machinery n=5

Systems

63.9 ms · H=16 · K=4 · L40

Pretrain

~50k H100-h · BS 16384

Neural

88h → 827h I2V · Defer factory

Empiricist: never quote Partial one-task smoke as Table 2 average

GR00T N1 vs π_0 / $\pi_{0.5}$ · same CFM family, different handoff

BEHAVIOR’26 ships BOTH as official baselines — dual-track is first-class

AXIS	GR00T N1	π_0 / $\pi_{0.5}$
Coupling	Separate DiT · cross-attn	Action expert · shared self-attn
VLM	Eagle-2 (SmolLM2+SigLIP-2)	PaliGemma lineage
Chunk / denoise	H=16 · K=4 · 10/120 Hz	H=50 (π_0) · ~10 Euler · 20–50 Hz
Embodiment I/O	Per-emb MLPs · LAPA/IDM	Pad/mask shared dim
Data bet	Pyramid + neural I2V	OXE-scale + $\pi_{0.5}$ co-train
Successor	N1.7 ~3B · H≤40 · Apache-2.0	$\pi_{0.5}$ + RTC + BEHAVIOR forks

FLAG: modular dual-system / cross-attn DiT vs fused shared-attn MoE expert · same chunk+flow interface

Verdict

Strong open humanoid VLA systems paper

Architectural foil to the π MoE coupling line

Core empirical wins over DP/BC-T look real;
cross-VLA SOTA claims and some aggregates need caution

MUST SAY

Preserve both DexMG numbers
(Table 2: 66.5 · Table 4: 58.5)

DO NOT

“Fix” the table mismatch
or silently pick one number

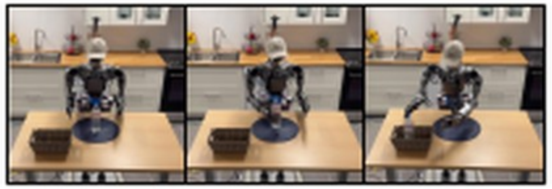
bottom shelf

Pick-and-Place with Left-to-Right Handover

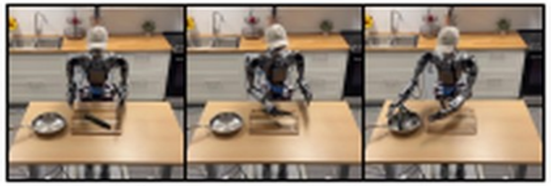
Pick up Novel Object to Novel Container

Post-Training Evaluations

Pick-and-Place: Tray to Plate



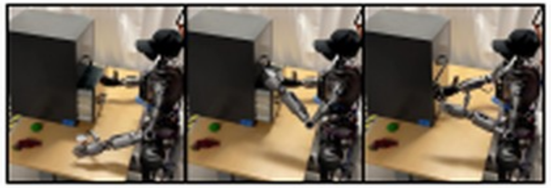
Pick-and-Place: Placemat to Basket



Pick-and-Place: Cutting Board to Pan

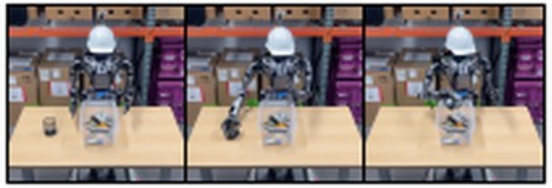
Articulated: White Drawer

Articulated: Wooden Chest

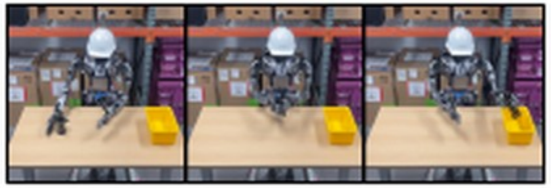


Articulated: Dark Cabinet

Industrial: Machinery Packing

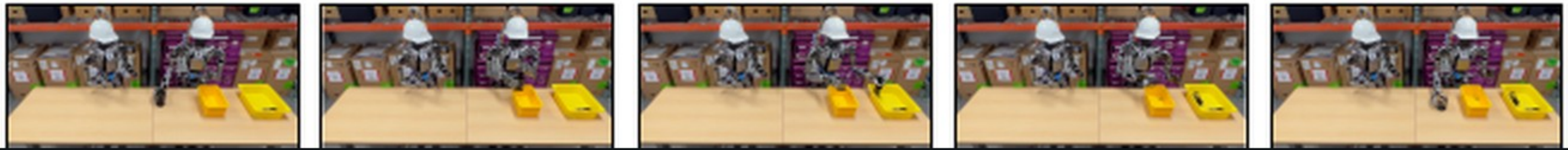


Industrial: Mesh Cup Pouring

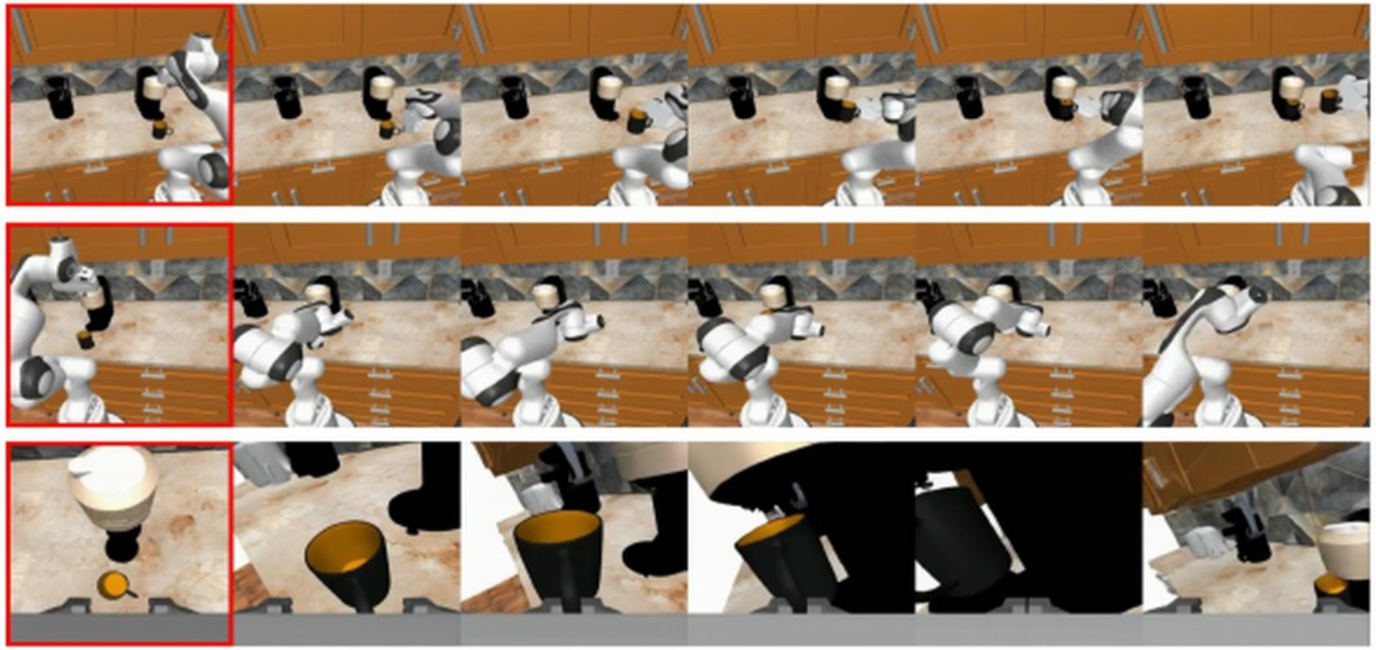


Industrial : Cylinder Handover

Coordination Part 1: Cylinder to Mesh Cup and Mesh Cup Handover to Another Robot



Prompt: `pick the mug from the counter and place it under the coffee machine dispenser, from left, right and wrist view`



Prompt (round 1): `pick up the green pepper`

Prompt (round 2): `move the green pepper to the bag`

Reviewer verdict

Strong systems paper · foil to π MoE coupling

Preserve both DexMG numbers (66.5 / 58.5)
Core DP/BC wins look real; VLA-SOTA needs caution

GR00T N1 · Oct 7 · all seats empty · still full five packs

VERSION DISCIPLINE · N1 ≠ N1.7

APIs / horizons / backbones differ — pin the checkpoint you serve

Paper N1

GR00T-N1-2B · ~2.2B (1.34B VLM)

Eagle-2 · SigLIP-2 + SmolLM2

H=16 · K=4 · mid-layer 12

63.9 ms / chunk @ L40 bf16

Table 2/3 tattoo numbers live HERE

N1.7 GA

nvidia/GR00T-N1.7-3B · ~3B

Cosmos-Reason2-2B (Qwen3-VL)

H ≤ 40 · new package / API

Apache-2.0 · BEHAVIOR’26 baseline

Do NOT quote N1 tables as N1.7

Stakeholder / Visionary rule

Prefer finetune N1.7 for product/challenge · use N1-2B only for paper replication
Match served horizon to trained horizon · N1.6 BEHAVIOR1k ≠ N1.7 forecast

Also: N1 is NOT a classical planner — generative CFM chunk controller with dual-system packaging